

Selection Operation : Algorithm 1: Linear Search.

Linear Search.

Roll_No	Name	Dept.	Year
1	Tasmia	CSE	3
2	Anisha	CSE	3
3	Tamima	EEE	3
4	Ibrahim	Pharmacy	2

now select the student whose Roll No = 3.

Algorithm A1 (Linear search)

- 1) Start from the 1st block of the file sequentially
- 2) Read each record one by one
- 3) Compare the attribute value with the condition
(Roll-No = 3)
- 4) Stop when the matching record is found.

Total cost = Number of block transfers +
Number of seeks.

If no condition on key ^{attribute} (Need to scan all)

Scan all blocks

so cost = $brc + 1 \text{ seek}$.

If the condition is on a key attribute,
only one record

example dept CSE is not the
key attribute, we get multiple records for
this.

(27/3) sequential access $\Rightarrow 1 \text{ seek}$, after
further visited record further ~~one~~
visit $\Rightarrow 1 \text{ seek}$.

For key attribute record can be found in:

→ In the first block

→ In the last block,

→ Anywhere in between.

so the average number of blocks read

$$= \frac{brc}{2}$$

$$\text{Average cost} = \frac{brc}{2} + 1 \text{ seek.}$$

Clustering Index

Student Table

	Roll-No	Name	Dept	Year
Record 1	1	Tasnia	CSE	3
Record 2	2	Anisha	CSE	2
Record 3	3	Maria	EEE	3
Record 4	4	Ibrahim	Pharmacy	2
Record 5	5	Helal	BBA	2

→ File has 5 blocks

→ We want to retrieve Roll-No = 3

→ There is a clustering index on Roll-No stored as a B+ tree.

→ Index height, $h_i = 2$ (root → leaf)

→ Seek time, $A_s = 5 \text{ ms}$

→ Block transfer time, $A_t = 2 \text{ ms}$

Traverse the index tree to find the leaf node containing Roll-No = 3.

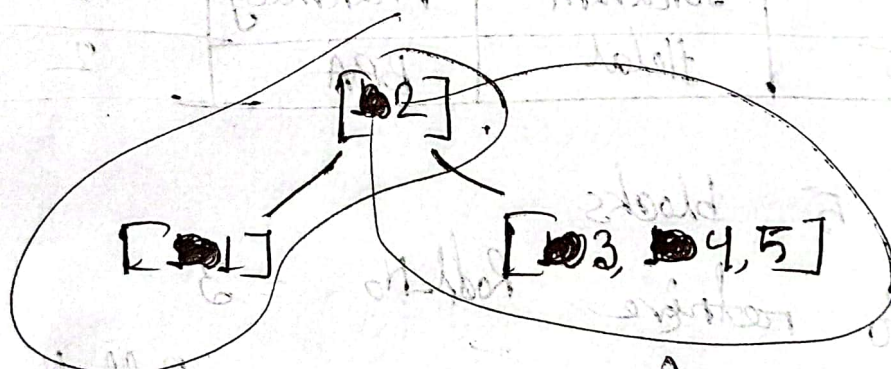
Root $\rightarrow$ Internal Node $\rightarrow$ Leaf Node

Cost = $h_i \times (t_T + t_S)$ cost

Retrieve the data block from disk to memory containing Roll_NO = 3.

Cost = $(t_T + t_S)$.

Total cost = $(h_i + 1) \cdot (t_T + t_S)$.



For left node

cost $(t_T + t_S)$

For right node

cost $(t_T + t_S)$

for multiple set record

Cost: $h_i \cdot (t_T + t_S) + t_S + t_T \cdot b$

Non Clustering Selection/ Secondary Index Search

Case 1: Candidate key $\neq$ Roll-No

Similar to clustering Index

Case 2: ~~not~~ a candidate key.

Suppose 4 records matches

Those 4 records are stored in a different block.

Cost $hi(ts + t_f) + nts + ntf$

Records are not ordered.

Clustering Index (Comparison)

$\sigma A \geq v \rightarrow$ use the index to find first record $\geq v$
then scan sequentially from there.
example:

first record ≥ 4 (Roll-No)

scan sequentially

4 $\rightarrow$ 5

~~Index traversal~~ to leaf hi

Traverse index tree to find the first record $\geq v$
 $= hi(ts + t_f)$

Scan sequentially

$$= b \cdot n (t_T + t_s)$$

$$\text{Cost} = (h_i + b_s) (t_s + t_T)$$

Conjunctive Selection

* select *

from student

where Dept = 'CSE' and Year = 2

for Dept ^{fetch record} $\Rightarrow$ Record 1, Record 2.

for Year ^{fetch record} $\Rightarrow$ Record 2.

$$\text{Cost} = h_i (t_s + t_T) + n * (t_s + t_T)$$

For index
traverse

To fetch actual
data blocks.

Conjunctive Selection

directly points to all records where
Dept = 'CSE' and Year = 2.

$$\text{Cost} = h_i \cdot (t_s + t_r) + n(t_s + t_r)$$

Conjunctive Selection by intersection of identifications

Dept = 'CSE' $\rightarrow$ RIDs = {101, 103, 105, 106}

Year = 2 $\rightarrow$ RIDs = {101, 103, 106}

Intersection = {101, 103, 106}

Only records satisfying both conditions are fetched.

secondary index on Dept $\rightarrow$ points to record pointers (RIDs)

Secondary index on Year = Points to record pointers (RIDs).

$$\begin{aligned} \text{Cost} &= \text{traverse 1st index} + \text{traverse second index} + \text{fetch } n \text{ records} \\ &= h_i(t_r + t_s) + h_i(t_r + t_s) + n(t_r + t_s) \end{aligned}$$

Disjunctive Selection

Secondary index on Dept $\rightarrow$ Points to RIDs

Secondary index on Year $\rightarrow$ Points to RIDs

Select *
from student

where Dept = 'CSE' or Year = 2;

Dept = 'CSE' $\rightarrow$ RIDs $\{101, 102\}$

Year = 2 $\rightarrow$ RIDs $\{102, 104, 105\}$

all $\{101, 102, 104, 105\}$ records are retrieved:

Cost = Index traversal for dept +
Index traversal for Year +

fetch n records